Sección A: Métodos numéricos

Sumas de Riemann

1) $F(x) = x^2$ en $[0,2]$, $n=4$

$\Delta x = 0.5$

x: 0, 0.5, 1, 1.5, 2

F: 0, 0.25, 1, 2.25, 4

$L4 = 0.5(0 + 0.25 + 1 + 2.25) = 1.75$

$R4 = 0.5(0.25 + 1 + 2.5 + 4) = 3.75$

$\bar{x}$: 0.25, 0.75, 1.25, 1.75 → F = 0.0625, 0.5625, 1.5625, 3.0625

$M4 = 0.5(5.25) = 2.625$

$\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3 \approx 2.6667$

$L4 = 1.75$ $R4 = 3.75$ $M4 = 2.625$

2) $\int_1^3 (1/x)\,dx$, $n=5$

$\Delta x = 0.4$

x: 1, 1.4, 1.8, 2.2, 2.6, 3.0

F: 1, 0.7143, 0.5556, 0.4545, 0.3846, 0.3333

$L5 = 0.4(1 + 0.7143 + 0.5556 + 0.4545 + 0.3846) = 0.4(3.109) \approx 1.2436$

$R5 = 0.4(0.7143 + 0.5556 + 0.4545 + 0.3846 + 0.3333) = 0.4(2.4423) \approx 0.9769$

$\bar{x}$: 1.2, 1.6, 2.0, 2.4, 2.8 → F = 0.8333, 0.625, 0.5, 0.4167, 0.3571

$M5 = 0.4(2.7321) \approx 1.0929$ $\int_1^3 dx/x = \ln 3 \approx 1.0986$

$L5 \approx 1.2436$ $R5 \approx 0.9769$ $M5 \approx 1.0929$

3) $F(x) = \sqrt{x}$ en $[0,4]$, $n=4$

$\Delta x = 1$

x: 0, 1, 2, 3, 4

F: 0, 1, 1.4142, 1.7321, 2

$L4 = 1(0+1+1.4142+1.7321) = 4.1463$

$R4 = 1(1+1.4142+1.7321+2) = 6.1463$

$\bar{x}$: 0.5, 1.5, 2.5, 3.5 → F: 0.7071, 1.2247, 1.5811, 1.8708

$M4 = 5.3838$

$\int_0^4 \sqrt{x}\,dx = [2x^{(3/2)}/3]_0^4 = 16/3 \approx 5.3333$

$L4 \approx 4.1463$ $R4 \approx 6.1463$ $M4 \approx 5.3838$

4) $F(x) = e^x$ en $[0,1]$, $n=6$

$\Delta x = 1/6 \approx 0.1667$

x: 0, 1/6, 2/6, 3/6, 4/6, 5/6, 1

F: 1, 1.1814, 1.3956, 1.6487, 1.9477, 2.3010, 2.7183

$L6 = 0.1667(1+1.1814+1.3956+1.6487+1.9477+2.3010)$
$= 0.1667(9.4744) \approx 1.5791$

$R6 = 0.1667(1.1814+1.3956+1.6487+1.9477+2.3010+2.7183) = 0.1667(11.1927) \approx 1.8654$

$\bar{x}$: 1/12, 3/12, 5/12, 7/12, 9/12, 11/12 → $F \approx$ 1.0869, 1.2840, 1.5169, 1.7920, 2.1170, 2.5009

$M6 = 0.1667(10.2978) \approx 1.7163$

$\int_0^1 e^x\,dx = e-1 \approx 1.7183$

$L6 \approx 1.5791$ $R6 \approx 1.8654$ $M6 \approx 1.7163$

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5) $\int_0^{\pi} \operatorname{sen} x\,dx$, $n=4$

$\Delta x = \pi/4$

x: $0, \pi/4, \pi/2, 3\pi/4, \pi$

F: $0, 0.7071, 1, 0.7071, 0$

$L4 = (\pi/4)(0+0.7071+1+0.7071) = (\pi/4)(2.4142) \approx 1.8961$

$R4 = (\pi/4)(0.7071+1+0.7071+0) = (\pi/4)(2.4142) \approx 1.8961$

$\bar{x}$: $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8 \rightarrow F = 0.3827, 0.9239, 0.9239, 0.3827$

$M4 = (\pi/4)(2.6131) \approx 2.0523$

$\int_0^{\pi} \operatorname{sen} x\,dx = [-\cos x]_0^{\pi} = 2$

$L4 = R4 \approx 1.8961$ $M4 \approx 2.0523$

Regla del trapecio

7) $\int_0^2 x^2\,dx$, $T4$

$\Delta x = 0.5$

x: $0, 0.5, 1, 1.5, 2 \rightarrow F: 0, 0.25, 1, 2.25, 4$

$T4 = (0.5/2)[0+4+2(0.25+1+2.25)] = 0.25[4+7] = 0.25(11)$

$T4 = 2.75$

2) $\int_1^3 (1/x)\,dx$, $T5$

$\Delta x = 0.4$

x: $1, 1.4, 1.8, 2.2, 2.6, 3 \rightarrow F: 1, 0.7143, 0.5556, 0.4545, 0.3846, 0.3333$

$T5 = (0.4/2)[1+0.3333+2(0.7143+0.5556+0.4545+0.3846)]$

$= 0.2[1.3333+2(2.1090)] = 0.2(5.5513)$

$T5 \approx 1.1103$

3) $\int_0^1 e^x\,dx$, T6

$\Delta x = 1/6$

F: 1, 1.1814, 1.3956, 1.6487, 1.9477, 2.3010, 2.7183

$T6 = (1/12)[1 + 2.7183 + 2(1.1814 + 1.3956 + 1.6487 + 1.9477 + 2.3010)]$

$= (1/12)[3.7183 + 2(8.4744)] = (1/12)(20.6671)$

$T6 \approx 1.7223$

4) $\int_0^{(\pi/2)} \operatorname{Sen} x\,dx$, T4

$\Delta x = \pi/8$

x: $0, \pi/8, \pi/4, 3\pi/8, \pi/2$ → F: 0, 0.3827, 0.7071, 0.9239, 1

$T4 = (\pi/16)[0 + 1 + 2(0.3827 + 0.7071 + 0.9239)] = (\pi/16)(5.0274)$

$T4 \approx 0.9871$

5) $\int_1^2 \sqrt{x}\,dx$, T4

$\Delta x = 0.25$

x: 1, 1.25, 1.5, 1.75, 2 → F: 1, 1.1180, 1.2247, 1.3229, 1.4142

$T4 = (0.25/2)[2.4142 + 2(1.1180 + 1.2247 + 1.3229)] = 0.125(9.7454)$

$T4 \approx 1.2182$

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Regla del Punto medio

1) $\int_0^2 x^2\,dx$, M4

$\Delta x = 0.5$, $\bar{x}$: 0.25, 0.75, 1.25, 1.75 → F = 0.0625, 0.5625, 1.5625, 3.0625

M4 = 0.5 (5.25) = 2.625

M4 = 2.625

2) $\int_1^3 (1/x)\,dx$ M5

$\Delta x = 0.4$, $\bar{x}$: 1.2, 1.6, 2.0, 2.4, 2.8 → F: 0.8333, 0.625, 0.5, 0.4167, 0.3571

M5 = 0.4 (2.7321) ≈ 1.0929

M5 ≈ 1.0929

3) $\int_0^1 e^x\,dx$, M6

$\Delta x = 1/6$, $\bar{x}$: 1/12, 3/12, 5/12, 7/12, 9/12, 11/12 → F ≈ 1.0869, 1.2840, 1.2840, 1.5169, 1.7960, 2.1170, 2.5009

M6 = (1/6) (10.2978) ≈ 1.7163

M6 ≈ 1.7163

4) $\int_0^{(\pi/2)} \cos x\,dx$, M4

$\Delta x = \pi/8$, $\bar{x}$: $\pi/16$, $3\pi/16$, $5\pi/16$, $7\pi/16$

F = $\cos(\pi/16) = 0.9808$, $\cos(3\pi/16) = 0.8315$, $\cos(5\pi/16) = 0.5556$, $\cos(7\pi/16) = 0.1951$

M4 = $(\pi/8)(2.5629) \approx 1.0065$

M4 ≈ 1.0065

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5) $\int_1^2 \ln x \, dx$, M4

$\Delta x = 0.25$ $\bar{x}$: 1.125, 1.375, 1.625, 1.875

$F = \ln(1.125) = 0.1178$, $\ln(1.375) = 0.3185$, $\ln(1.625)$
$= 0.4855$, $\ln(1.875) = 0.6286$

$M4 = 0.25(1.5504) \approx 0.3876$

$M4 \approx 0.3876$

IV Regla de Simpson

1) $\int_0^2 x^2 \, dx$, S4

$\Delta x = 0.5$ x: 0, 0.5, 1, 1.5, 2 → F: 0, 0.25, 1, 2.25, 4

$S4 = (0.5/3)[0+4+4(0.25)+2(1)] = (1/6)[4+10+2] = 16/6$

$S4 = 8/3 \approx 2.6667$

2) $\int_1^3 (1/x) \, dx$, S4

$\Delta x = 0.5$, x: 1, 1.5, 2, 2.5, 3 → F: 1, 0.6667, 0.5, 0.4, 0.3333

$S4 = (0.5/3)[1+0.3333+4(0.6667+0.4)+2(0.5)] = (1/6)(6$

$S4 = 1.1000$

3) $\int_0^1 e^x \, dx$, S6

$\Delta x = 1/6$, F: 1, 1.1814, 1.3956, 1.6487, 1.9477, 2.3010, 2.7183

$S6 = (1/18)[1+2.7183+4(1.1814+1.6487+2.3010)+2$
$(1.3956+1.9477)]$

$= (1/18)[3.7183+20.5244+6.6866] = (1/18)(30.9293)$

$S6 \approx 1.7183$

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4) $\int_0^{\pi/4} \sqrt{(1+x^3)}\,dx$, S4

$\Delta x = \pi/4$, F: 0, $\sqrt{2}/2$, 1, $\sqrt{2}/2$, 0

$S4 = (\pi/12)[0+0+4\sqrt{2}+2(1)] = (\pi/12)(4\sqrt{2}+2) = \pi(1+2\sqrt{2})/6$

$S4 \approx 2.0046$

5) $\int_0^{1} \sqrt{(1+x^3)}\,dx$, S4

$\Delta x = 0.25$, F: 1, 1.00778, 1.06066, 1.19242, 1.41421

$S4 = (1/12)[1+1.41421+4(1.00778+1.92242)+2(1.06066)]$

$= (1/12)[2.41421+8.8008+2.12132] = (1/12)(13.33633)$

$S4 = 1.1114$

V Integral definida y Area

1) $\int_0^2 (3x^2-2x+1)\,dx$

$F(x) = x^3-x^2+x$

$F(2) = 8-4+2 = 6$, $F(0) = 0$

$= 6$

2) $\int_1^e (1/x)\,dx$

$F(x) = \ln x$

$F(e) - F(1) = 1-0$

$= 1$

3) $\int_0^{\pi/2} \cos x\,dx$

$F(x) = \operatorname{Sen} x$

$F(\pi/2) - F(0) = 1-0$

$= 1$

4) Area entre $F(x) = x^2$ y $g(x) = 2x$

$x^2 = 2x \rightarrow x^2-2x = 0 \rightarrow x(x-2) = 0 \rightarrow x = 0, x = 2$

En $(0,1)$: $g(x) = 2x > F(x) = x^2$ (ej. $x = 1$: $2 > 1$)

$\text{Area} = \int_0^2 (2x-x^2)\,dx = [x^2 - x^3/3]_0^2 = (4-8/3)-0$

$\text{Area} = 4/3\ u^2$

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5) Area bajo $F(x) = e^x$, $x \in [0,1]$

$\text{Area} = \int_0^1 e^x \, dx = [e^x]_0^1 = e - 1$

$\text{Area} = e - 1 \approx 1.71828 \, u^2$

Seccion B: Integrales indefinidas

VI. Integracion directa

4) $\int (ax + by)^2 \, dx$

$(ax+by)^2 = a^2x^2 + 2abxy + b^2y^2$

$\int a^2x^2 \, dx = a^2x^3/3, \int 2abxy \, dx = abx^2y, \int b^2y^2 \, dx = b^2xy^2$

$F(x) = a^2x^3/3 + abx^2y + b^2xy^2 + C$

2) $\int (3/x^2) - 9/\sqrt{x}) \, dx$

$3x^{-2} - 9x^{\wedge}(-1/2)$

$\int 3x^{-2} \, dx = -3/x, \int 9x^{\wedge}(-1/2) \, dx = 18\sqrt{x}$

$F(x) = -3/x - 18\sqrt{x} + C$

3) $\int (x + 4\sqrt{x} - 4/x) \, dx$

$\int x \, dx = x^2/2, \int 4\sqrt{x} \, dx = (8/3)x^{\wedge}(3/2), \int 4/x \, dx = 4\ln|x|$

$F(x) = x^2/2 + (8/3)x^{\wedge}(3/2) - 4\ln|x| + C$

4) $\int (x - 1 - 1/x) \, dx$

$\int x \, dx = x^2/2, \int 1 \, dx = x, \int 1/x \, dx = \ln|x|$

$F(x) = x^2/2 - x - \ln|x| + C$

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5) $\int (x-1)(x+1)\,dx$

$(x-1)(x+1) = x^2 - 1$

$\int (x^2-1)\,dx = x^3/3 - x$

VII. Sustitucion (Potencias)

4) $\int x^2/\sqrt{(x^3+2)}\,dx$

$u = x^3+2, \quad du = 3x^2\,dx \rightarrow x^2\,dx = du/3$

$\int u^{(-1/2)}/3\,du = (1/3)(2\sqrt{u}) = (2/3)\sqrt{u}$

$F(x) = (2/3)\sqrt{(x^3+2)} + C$

2) $\int (x-2)/(1-2x)^4\,dx$

$u = 1-2x, \quad x = (1-u)/2, \quad dx = -du/2, \quad x-2 = (-3-u)/2$

$\int [(-3-u)/2]/u^4\,(-du/2) = (1/4)\int (3+u)u^{-4}\,du = (1/4)$

$(-u^{-3} - u^{-2}/2)$

$= -u^{-3}/4 - u^{-2}/8$, en terminos de $(2x-1)$

$F(x) = -1/[8(2x-1)^2] + 1/[4(2x-1)^3] + C$

3) $\int x/\sqrt{(x^2+2x-4)}\,dx$

$x = (1/2)(2x+2) - 1$

$(1/2)\int (2x+2)(x^2+2x-4)^{(-1/2)}\,dx = \sqrt{(x^2+2x-4)}$

$x^2+2x-4 = (x+1)^2 - 5 \rightarrow -\int dx\,\sqrt{((x+1)^2-5)} = -\ln|x+1+\sqrt{(x^2+2x-4)}|$

$F(x) = \sqrt{(x^2+2x+4)} - \ln|x+1+\sqrt{(x^2+2x-4)}| + C$

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4) $\int \sqrt{x}/(1+\sqrt{x})^3\,dx$

$t=\sqrt{x},\ x=t^2,\ dx=2t\,dt \to 2\int t^2/(1+t)^3\,dt$

$w=1+t: 2\int (w^2-2w+1)/w^3\,dw = 2\int (1/w - 2/w^2 + 1/w^3)\,dw$

$= 2\ln|w| + 4/w - 1/w^2,\ w=1+\sqrt{x}$

$F(x)= 2\ln(1+\sqrt{x}) + 4/(1+\sqrt{x}) - 1/(1+\sqrt{x})^2 + C$

5) $\int (x+3)\sqrt{(x^2+6x)}\,dx$

$u=x^2+6x,\ du=2(x+3)\,dx \to (x+3)\,dx = du/2$

$(1/2)\int u\,du = (1/2)(2/3)u^{\wedge}(3/2) = (1/3)u^{\wedge}(3/2)$

$F(x)= (1/3)(x^2+6x)^{\wedge}(3/2) + C$

VIII. Exponenciales

1) $\int e^x/(e^{\wedge}(2x)+3)\,dx$

$u=e^x,\ du=e^x\,dx$

$\int du/(u^2+3) = (1/\sqrt{3})\arctan(u/\sqrt{3})$

$F(x) = (1/\sqrt{3})\arctan(e^x/\sqrt{3}) + C$

2) $\int dx/(e^x+e^{-x})$

de arriba y abajo: $e^x/(e^{\wedge}(2x)+1)$

$u=e^x: \int du/(u^2+1) = \arctan$

$F(x) = \arctan(e^x) + C$

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3) $\int 5^x e^x \, dx$

$5^x e^x = (5e)^x$

$\int (5e)^x \, dx = (5e)^x / \ln(5e) = (5e)^x / (1 + \ln 5)$

$F(x) = 5^x e^x / (1 + \ln 5) + C$

4) $\int (e^{\wedge}(2x) + 3e^x) \, dx$

$(a^x - b^x)^2 = a^{\wedge}(2x) - 2(ab)^x + b^{\wedge}(2x)$

5) $\int a^{\wedge}(2x) \, dx = a^{\wedge}(2x) / (2 \ln a)$, $\int (ab)^x \, dx = (ab)^x / (\ln$

$+ \ln b)$, $\int b^{\wedge}(2x) \, dx = b^{\wedge}(2x) / (2 \ln b)$

$F(x) = a^{\wedge}(2x) / (2 \ln a) - 2(ab)^x / (\ln a + \ln b) + b^{\wedge}(2x) /$
$(2 \ln b) + C$

IX. logaritmicas

1) $\int dx / [x \sqrt{(4 - \ln^2 x)}]$

$u = \ln x$, $du = dx/x$

$\int du / \sqrt{(4 - u^2)} = \arcsin(u/2)$

$F(x) = \arcsin(\ln x / 2) + C$

2) $\int dx / [x (\ln x)^2]$

$u = \ln x \Rightarrow \int u^{-2} \, du = -1/u$

$F(x) = -1/\ln x + C$

3) $\int dx / (x \ln x)$

$u = \ln x \Rightarrow \int du/u = \ln|u|$

$F(x) = \ln|\ln x| + C$

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4) $\int (x+2)/(2x^2+2x+1)\,dx$

$x+2 = (1/4)(4x+2)+3/2$

$(1/4)\int (4x+2)/(2x^2+2x+1)\,dx = (1/4)\ln(2x^2+2x+1)$

$2x^2+2x+1 = 2(x+1/2)^2+1/2 \Rightarrow (3/2)\int dx/(\dots) =$
$(3/2)\arctan(2x+1)$

$F(x) = (1/4)\ln(2x^2+2x+1)+(3/2)\arctan(2x+1)+C$

5) $\int dx/(x^2+2x+2)$

$x^2+2x+2 = (x+1)^2+1$
$\int du/(u^2+1) = \arctan u$

X. trigonometricas

1) $\int \sec^2 y/\cos y\,dy$

$\sec^2 y/\cos y = \sec^3 y$
Por partes sabido: $I = (1/2)[\sec y \tan y + \ln|\sec y+\tan y|]$

$F(y) = (1/2)[\sec y \tan y + \ln|\sec y+\tan y|]+C$

2) $\int dy/(1+\cos^2 y)$

$1/\cos^2 y = \sec^2 y/(\tan^2 y+2)$
$u = \tan y$: $\int du/(u^2+2) = (1/\sqrt{2})\arctan(u/\sqrt{2})$

$F(y) = (1/\sqrt{2})\arctan(\tan y/\sqrt{2})+C$

3) $\int (\cos x - \operatorname{sen} x)^2 \, dx$

$(\cos x - \operatorname{sen} x)^2 = 1 - \operatorname{sen}^2 x$

$\int (1 - \operatorname{sen} 2x) \, dx = x + (1/2) \cos 2x$

$F(x) = x + (1/2) \cos 2x + C$

4) $\int \operatorname{sen} 3x / \cos^4 3x \, dx$

$u = \cos 3x, \; du = 3 \operatorname{sen} 3x \, dx$

$-(1/3) \int u^{-4} \, du = 1/(9u^3)$

$F(x) = 1/[9 \cos^3 3x] + C$

5) $\int dx / [\cos^2 x \sqrt{(\tan x - 1)}]$

$u = \tan x - 1, \; du = \sec^2 x \, dx$

$\int u^{(-1/2)} \, du = 2\sqrt{u}$

$F(x) = 2\sqrt{(\tan x - 1)} + C$

X1. Trigonometria Inversas

1) $\int (3+x) / \sqrt{(1-x^2)} \, dx$

$3 \int dx / \sqrt{(1-x^2)} = 3 \operatorname{arc\,sin} x$

$u = 1 - x^2 : \int x / \sqrt{(1-x^2)} \, dx = -\sqrt{(1-x^2)}$

$F(x) = 3 \operatorname{arc\,sin} x - \sqrt{(1-x^2)} + C$

2.) $\int (2x-7)/(x^2+9)\,dx$

$\int 2x/(x^2+9)\,dx = \ln(x^2+9)$

$-7\int dx/(x^2+9) = (-7/3)\arctan(x/3)$

$F(x) = \ln(x^2+9) - (7/3)\arctan(x/$

3) $\int dx/[x\sqrt{(4-9\ln^3 x)}]$

$u = \ln x : \int du/\sqrt{(4-3u^2)}$

$= (1/3)\arcsin(3u/2)$

$F(x) = (1/3)\arcsin(3\ln x/2) + C$

4.) $\int 2e^{(2x)}/(7+e^{(4x)})\,dx$

$u = e^{(2x)}, \quad du = 2e^{(2x)}\,dx$

$\int du/(7+u^2) = \arctan u$

$F(x) = \arctan(e^{(2x)}) + C$

5) $\int 5\sec^2 x/\sqrt{(7-4\tan^2 x)}\,dx$

$u = \tan x : \int du/\sqrt{(7-4u^2)}$

$= (7/2)\arcsin(2u)$

XII. Hiperbolicas Inversas

1) $\int dx/\sqrt{(x^2+16)}$ Forma directa, $a=4$

$F(x)=\ln(x+\sqrt{(x^2+16)})+C$

2) $\int dx/\sqrt{(4x^2-25)}$ $\quad u=2x,\ dx=du/2$

$(1/2)\int du/\sqrt{(u^2-25)}$

$F(x)=(1/2)\ln(2x+\sqrt{(4x^2-25)})+C$

3) $\int (x+2)/\sqrt{(x^2+9)}\,dx$

$\int x/\sqrt{(x^2+9)}\,dx=\sqrt{(x^2+9)}$

$2\int dx/\sqrt{(x^2+9)}=2\ln(x+\sqrt{(x^2+9)})$

4) $\int dx/\sqrt{(b^2x^2-a^2)}$

$u=bx,\ dx=du/b$

$(1/b)\int du/\sqrt{(u^2-a^2)}$

$F(x)=(1/b)\ln(bx+\sqrt{(b^2x^2-a^2)})+C$

5) $\int dx/\sqrt{(x^2-25)}$

Forma directa, $a=5$

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XIII. Trinomio ax^2+bx+c

1) $\int (2x+3)/\sqrt{(9x^2-12x+9)}\,dx$

$2x+3=(7/9)(18x-12)+13/3$

$(1/9)(2\sqrt{(9x^2-12x+8)})=(2/9)\sqrt{(9x^2-12x+8)}$

$9x^2-12x+8=(3x-2)^2+4 \rightarrow (13/9)\ln(3x-2+\sqrt{(9x^2-12x+8)})$

$F(x)=(2/9)\sqrt{(9x^2-12x+8)}+(13/9)\ln(3x-2+\sqrt{(9x^2-12x+8)})+C$

2) $\int (x+2)/\sqrt{(4x-x^2)}\,dx$

$x+2=-(-1/2)(4-2x)+4$

$-(1/2)(2\sqrt{(4x-x^2)})=-\sqrt{(4x-x^2)}$

$4x-x^2=4-(x-2)^2 \rightarrow 4\arcsin((x-2)/2)$

$F(x)=-\sqrt{(4x-x^2)}+4\arcsin((x-2)/2)+C$

3) $\int (2x+2)/\sqrt{(x^2+2x-3)}\,dx$

$2x+2=$ derivada exacta del radicando

$\int F'/\sqrt{F}\,dx=2\sqrt{F}$

$F(x)=2\sqrt{(x^2+2x-3)}+C$

4) $\int (2x-1)/\sqrt{(4x^2+4x-3)}\,dx$

$2x-1=(1/4)(8x+4)-2$

$(1/4)(2\sqrt{(\ldots)})=(1/2)\sqrt{(4x^2+4x-3)}$

$4x^2+4x-3=(2x+1)^2-4 \rightarrow -\ln(2x+1+\sqrt{(4x^2+4x-3)})$

$F(x)=(1/2)\sqrt{(4x^2+4x-3)}-\ln(2x+1+\sqrt{(4x^2+4x-3)})+C$

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5) $\int dx / \sqrt{(2x^2+2x+5)}$

$2x^2+2x+5 = 2(x+1/2)^2+9/2$

$(1/2)\ln(x+1/2+\sqrt{(x^2+x+9/4)})$

$=(\sqrt{2}/2)\ln(2x+1+\sqrt{4x^2+4x+10}))$

$F(x)=(\sqrt{2}/2)\ln(2x+1+\sqrt{(4x^2+4x+10)})+C$

XIV. Integracion Por Partes

1). $\int e^{(ax)}\cos(bx)\,dx$

$I=(e^{(ax)}/a)\cos(bx)+(b/a)\int e^{(ax)}\operatorname{sen}(bx)\,dx$

$J=(e^{(ax)}/a)\operatorname{sen}(bx)-(b/a)I$

$I=(e^{(ax)}/a)\cos(bx)+(b/a)J$

$F(x)=e^{(ax)}(a\cos bx+b\operatorname{sen}bx)/(a^2+b^2)+C$

2) $\int e^{(ax)}\operatorname{sen}(bx)\,dx$

Mismo ciclo, intercambiando seno/coseno

$F(x)=e^{(ax)}(a\operatorname{sen}bx-b\cos bx)/(a^2+b^2)+C$

3) $\int \operatorname{sen}(\ln x)\,dx$

$u=\operatorname{sen}(\ln x),\ dv=dx : I=x\operatorname{sen}(\ln x)-\int\cos(\ln x)$

$\int\cos(\ln x)\,dx = x\cos(\ln x)+1$

$I=x\operatorname{sen}(\ln x)-x\cos(\ln x)-I \rightarrow 2I=x[\operatorname{sen}(\ln x)-\cos(\ln x)]$

$F(x)=(x/2)[\operatorname{sen}(\ln x)-\cos(\ln x)]+C$

4.) $\int x \arctan x \, dx$

$u = \arctan x, \; du = x\,dx, \; v = x^2/2$

$I = (x^2/2)\arctan x - (1/2)\int x^2/(1+x^2)\,dx$

$x^2/(1+x^2) = 1 - 1/(1+x^2)$

$\rightarrow (1/2)[x - \arctan x]$

$I = (x^2/2)\arctan x - x/2 + (1/2)\arctan x$

$F(x) = (x^2+1)/2 \cdot \arctan x - x/2 + C$

5.) $\int \ln(x^2+1)\,dx$

$u = \ln(x^2+1), \; dv = dx, \; v = x$

$I = x\ln(x^2+1) - 2\int x^2/(x^2+1)\,dx$

$x^2/(x^2+1) = 1 - 1/(x^2+1) \rightarrow 2[x - \arctan x]$

$I = x\ln(x^2+1) - 2x + 2\arctan x$

$F(x) = x\ln(x^2+1) - 2x + 2\arctan x + C$